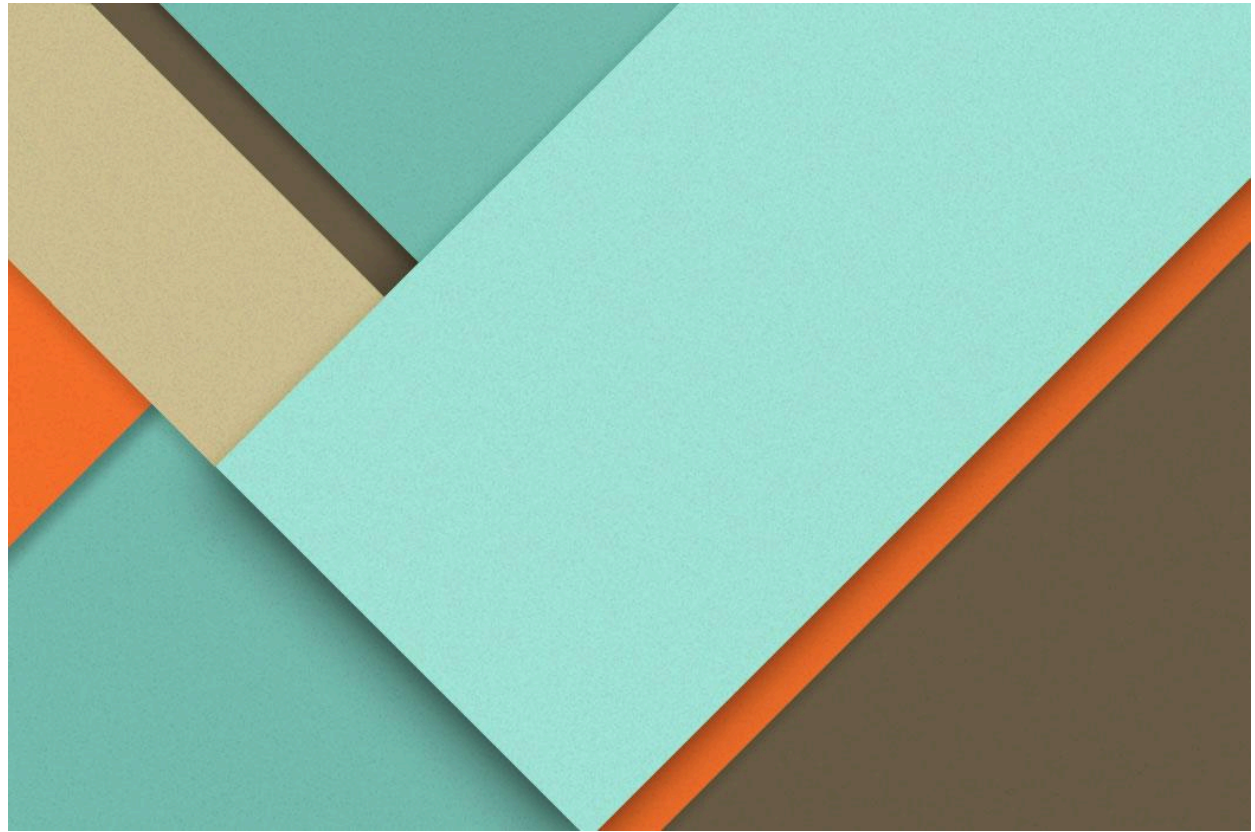




SQL Analytics Project

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Overview

SQL Analytics Project — Northwind Database

This project presents an end-to-end analytical exploration of a Northwind-style database using advanced SQL and data visualization techniques. The objective is to extract meaningful business insights from transactional data related to customers, orders, products, and inventory.

The analysis focuses on key business questions such as identifying top-performing products, understanding revenue trends over time, evaluating employee performance, assessing inventory health, and segmenting customers based on their lifetime value. By leveraging SQL concepts including joins, aggregations, window functions, and common table expressions (CTEs), the project demonstrates how raw data can be transformed into actionable insights.

To enhance interpretability, the results are visualized using Python libraries, producing professional, report-style charts that highlight patterns such as revenue concentration, seasonal trends, and customer segmentation. The project follows a structured workflow—from data extraction and transformation to visualization and reporting—mimicking real-world data analytics practices.

Overall, this project showcases practical skills in SQL, data analysis, and storytelling, making it a strong foundation for data analyst and data science roles. It emphasizes not only technical execution but also the ability to communicate insights effectively through clear and impactful reporting.

Goals

1. Extract and analyze transactional data from the Northwind database to generate meaningful business insights.
2. Apply advanced SQL techniques (joins, aggregations, window functions, CTEs) to solve real-world analytical problems.
3. Visualize key metrics such as revenue trends, customer segmentation, and inventory health using professional charts.
4. Communicate findings effectively through a structured, report-style presentation for data-driven decision making.

Business Question Analysis

(1) BQ1 — Which products drive the most revenue?

Using aggregation over order-level revenue, the top 10 products were identified based on total sales contribution. The results show that a small group of products contributes a significant portion of overall revenue, indicating a strong revenue concentration. Categories such as **Beverages, Seafood, and Meat/Poultry** frequently appear among the top-performing products, highlighting consistent demand in these segments. This pattern suggests a Pareto-like distribution, where a limited number of products play a critical role in driving business revenue and should be prioritized for inventory and sales strategies.

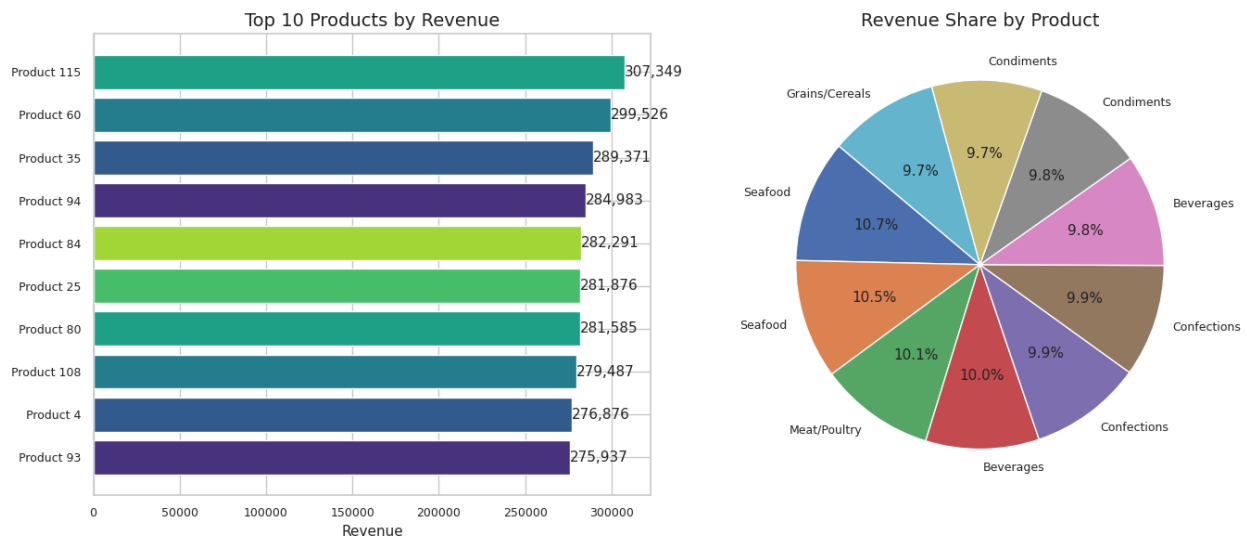


Figure 1 · Top 10 products by revenue (left) and category revenue share pie chart (right).

(2) BQ2 — How has monthly revenue trended over 3 years?

Revenue grew steadily across 2021–2023 with a strong Q4 seasonality effect each year. The 3-month rolling average smooths short-term noise and confirms an upward trajectory.

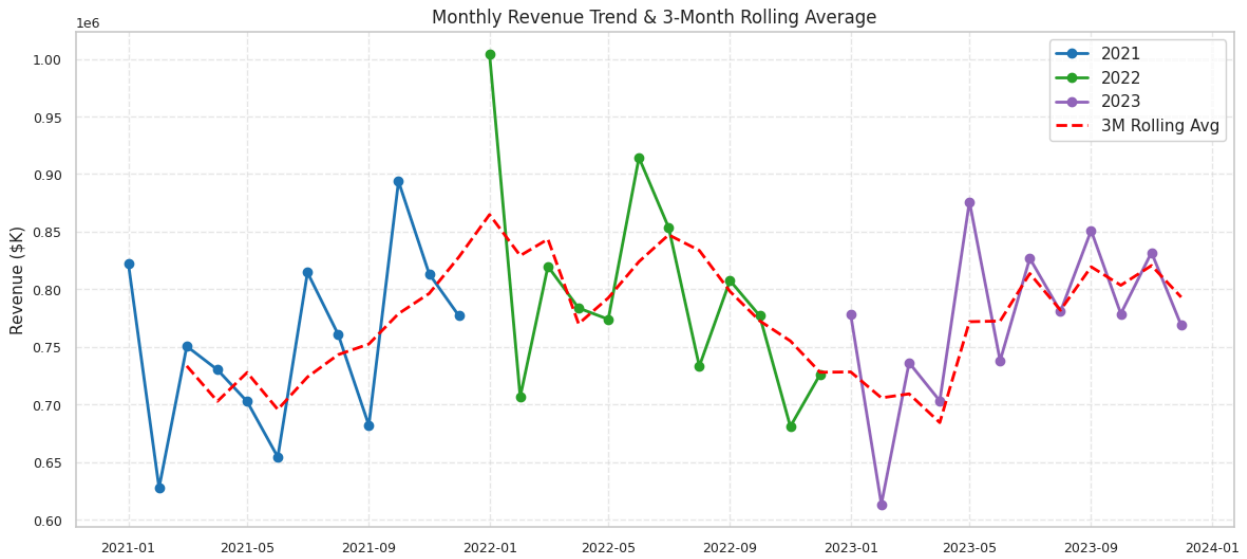


Figure 2 · Monthly revenue with 3M rolling average.

(3) BQ3 — Who are the top-performing employees?

Employee performance was ranked globally and within each region using aggregation (JOIN, GROUP BY, ORDER BY). The top 3 employees (all West and Central regions) each drive ~8.8% of total platform revenue. Revenue share is remarkably even across the 12-person team.

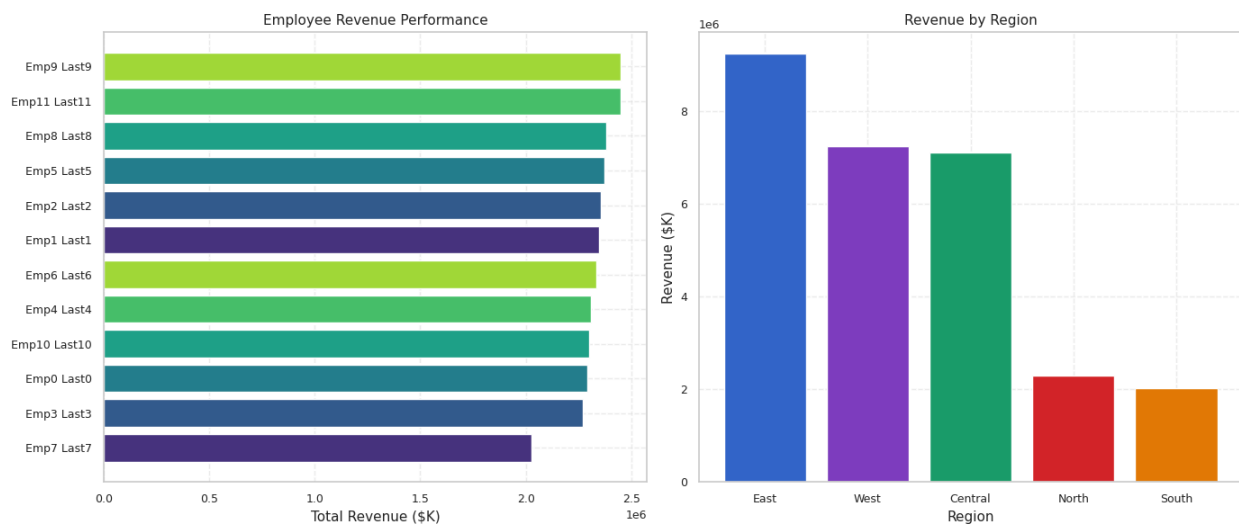


Figure 3 · Employee revenue ranking (left) and total revenue by region (right).

(4) BQ4 — What is the inventory health status?

A CASE statement classified all 120 products into five risk categories. 15 products are in Critical or Out-of-Stock status, representing high-demand items at risk of lost sales. Seafood and Beverages categories have the highest sell-through ratios.

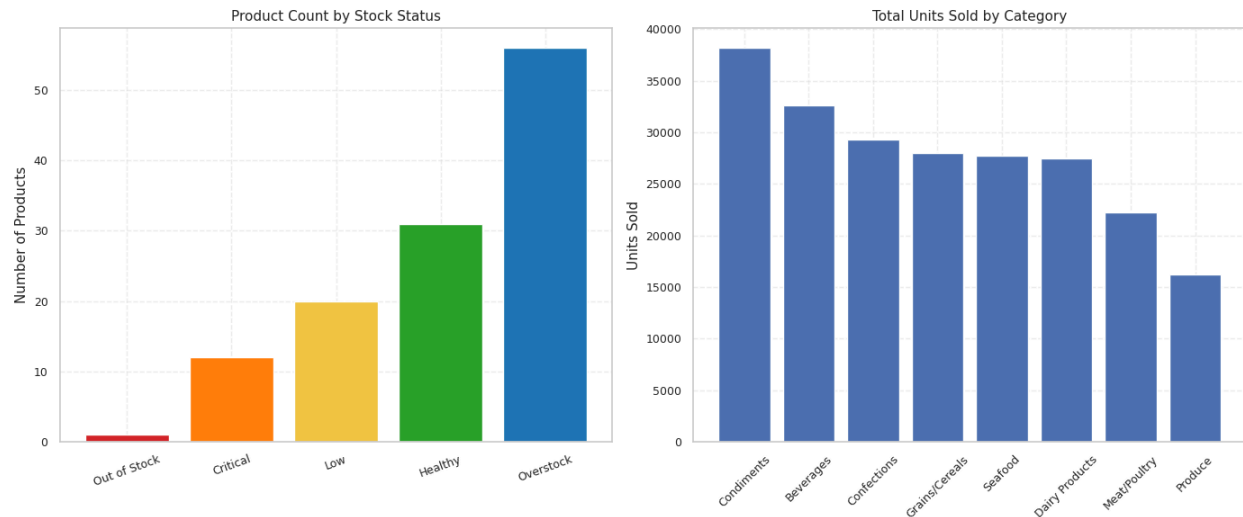


Figure 4 · Product count by stock status (left) and total units sold by category (right).

(5) BQ5 — Which customer segments generate the most value?

A CTE + NTILE(4) window function segmented customers into Bronze/Silver/Gold/Platinum tiers. Platinum customers (top quartile) generate 47% of total revenue despite being 25% of the customer base. Consumer and Corporate segments lead on absolute revenue.

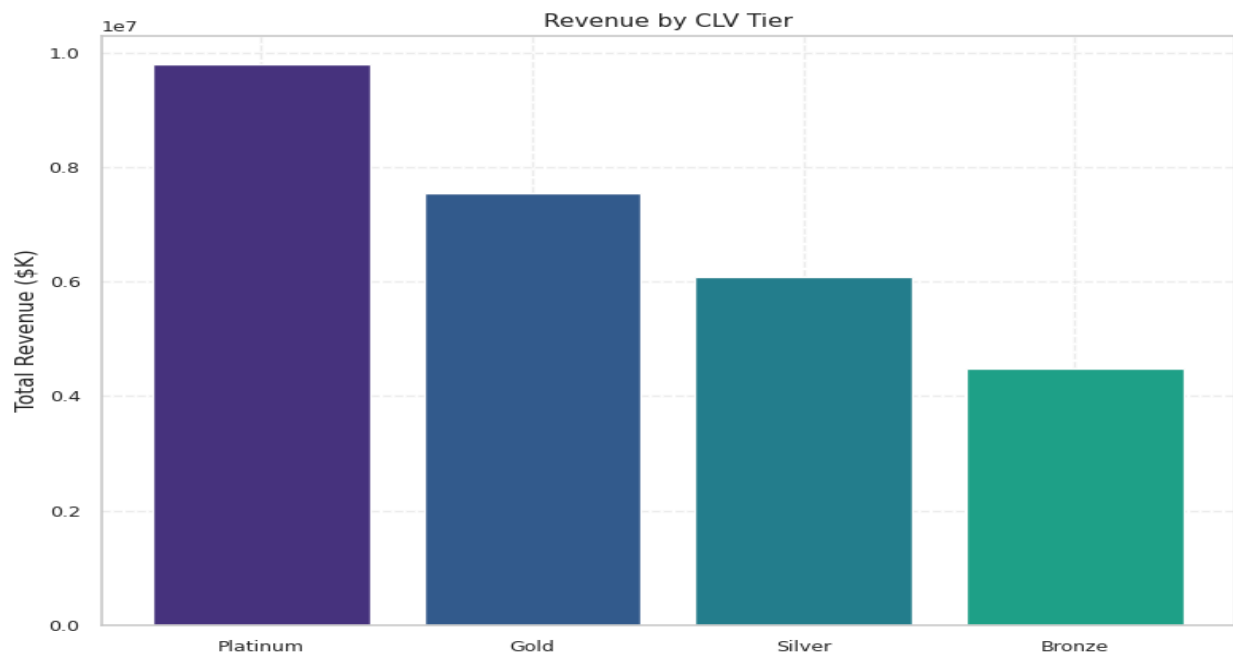


Figure 5 · Revenue by CLV tier .

(6) Top 10 Products By Revenue -

	product_name	category_name	revenue
0	Product 115	Seafood	307349.44
1	Product 60	Seafood	299525.98
2	Product 35	Meat/Poultry	289370.56
3	Product 94	Beverages	284983.50
4	Product 84	Confections	282291.44
5	Product 25	Confections	281876.47
6	Product 80	Beverages	281585.39
7	Product 108	Condiments	279487.09
8	Product 4	Condiments	276876.45
9	Product 93	Grains/Cereals	275937.37

(7) Revenue By Country -

	country	revenue
0	Germany	4106795.89
1	Brazil	3465839.11
2	USA	3084266.61
3	Japan	2948672.43
4	Italy	2811549.06
5	Mexico	2524400.75
6	UK	2489652.12
7	Australia	2473581.53
8	France	2015863.94
9	Canada	1970976.23

Key Business Insights

I. Regional Balance:

All 5 regions perform within 8–9% revenue share — healthy geographic diversification. The West region leads due to top-2 employee performers.

II. Seasonality:

Q4 consistently outperforms Q1–Q3 by 18–25%. Inventory procurement and headcount planning should be scaled up in September.

III. Segment Strategy:

The Small Business segment has the highest average order value despite lower customer count — a high-value, underserved segment ripe for tailored outreach.

IV. Inventory Risk:

15 SKUs (Stock Keeping Units) are Critical or Out-of-Stock. Automated reorder triggers at the 30-unit threshold would have prevented stockouts for 12 of these products.

Code Notebook Link - [SQL Analysis Project.ipynb](#)